DAYANANDA SAGAR COLLEGE OF ENGINEERING

(An Autonomous Institute affiliated to Visvesvaraya Technological University (VTU), Belagavi,
Approved by AICTE and UGC, Accredited by NAAC with 'A' grade)

Shavige Malleshwara Hills, Kumaraswamy Layout, Bengaluru - 560 111. India

Department of Information Science and Engineering

(Accredited by NBA Tier 1: 2025-2028)

A Project Synopsis

on

"Temporal Spiking Graph Convolutional Networks for Low Power Dynamic Transaction Fraud Detection"

Submitted as a part of the Final Year Project of

BACHELOR OF ENGINEERING

in

DEPARTMENT OF INFORMATION SCIENCE AND ENGINEERING

Submitted by

BHOOMIKA N	**1DS23IS031**
CHANDRA SHEKAR	**1DS23IS035**
CHETAN S	**1DS23IS037**
DIKSHITH M	**1DS23IS044**

Under the guidance of

Dr. Vaidehi M
Professor
Dept. of ISE, DSCE

2026-2027

<table>
<tr><td>TITLE OF THE PROJECT</td><td>Temporal Spiking Graph Convolutional Networks for Low-Power Dynamic Transaction Fraud Detection</td></tr>
<tr><td>PROJECT GUIDE DETAILS</td><td>Dr. Vaidehi M, Professor, Dept. Of ISE DSCE
PHONE NO.: +91 9449024707
EMAIL ID: vaidehim-ise@dayanandasagar.edu</td></tr>
<tr><td>STUDENT DETAILS</td><td>
<table>
<tr><td>NAME</td><td>USN</td><td>PHONE NO.</td><td>EMAIL ID</td></tr>
<tr><td>BHOOMIKA N</td><td>1DS23IS031</td><td>8050665900</td><td>1ds23is031@dsce.edu.in</td></tr>
<tr><td>CHANDRA SHEKAR</td><td>1DS23IS035</td><td>7975433944</td><td>1ds23is035@dsce.edu.in</td></tr>
<tr><td>CHETAN S</td><td>1DS23IS037</td><td>9590455200</td><td>1ds23is037@dsce.edu.in</td></tr>
<tr><td>DIKSHITH M</td><td>1DS23IS044</td><td>8971059191</td><td>1ds23is044@dsce.edu.in</td></tr>
</table>
</td></tr>
<tr><td>PROJECT TIMELINE (Tentative Start and End Date)</td><td>August 2026 - June 2027</td></tr>
<tr><td>PROJECT COORDINATORS</td><td>Dr. Santosh Anand, Prof. Ayush Mishra</td></tr>
<tr><td>DOMAIN OF THE PROJECT</td><td>Spiking Neural Networks (SNNs), Graph Neural Networks (GNNs) & Neuromorphic Engineering for Cyber-Security and Dynamic Network Anomaly Detection (Foundations of Algorithms in Artificial Intelligence & Deep Learning)</td></tr>
<tr><td>BACKGROUND OF PROJECT WITH REGARD TO THE</td><td>High Computational and Energy Cost of Conventional GNNs:</td></tr>
</table>

DRAWBACKS IN EXISTING PROJECTS	Conventional Graph Convolutional Networks (GCNs) rely on continuous numerical operations for feature aggregation and message passing across the graph. As the number of nodes and edges increases, these operations become computationally expensive, resulting in higher processing time and energy consumption, particularly for large and continuously changing graphs. • **Sensitivity to Short-Term Temporal Noise:** Conventional GNNs generally do not maintain an explicit temporal state across successive graph observations. As a result, short-lived changes or isolated noise in transaction and network activity may be interpreted as meaningful patterns, potentially increasing false-positive anomaly or fraud detections. • **Limited Use of Spike Sparsity for Energy-Efficient Computing:** Most existing graph-based anomaly detection methods are designed for conventional GPU or CPU-based systems and do not take advantage of the sparse, event-driven nature of Spiking Neural Networks (SNNs). This limits their potential for deployment on low-power neuromorphic and edge-computing platforms. • **Lack of Temporal and Event-Driven Neuron Dynamics:** Traditional GNN-based models primarily process graph features through continuous numerical activations and do not incorporate biologically inspired mechanisms such as membrane potential integration. In an SNN, the membrane potential accumulates incoming signals over time, as represented by . This temporal integration can help retain relevant information across successive time steps while reducing the effect of isolated fluctuations.
PROBLEM STATEMENT	Dynamic transaction and network graphs contain continuously changing relationships between entities, making the detection of fraudulent or anomalous activity challenging. Conventional Graph Neural Networks (GNNs) process graph information using continuous numerical operations, which can become computationally expensive as the size and activity of the graph increase. They also do not explicitly maintain temporal state in a conventional GCN setup, making them potentially sensitive to short-lived fluctuations in streaming data.

	This project aims to address these limitations by developing a Temporal Spiking Graph Convolutional Network (TS-GCN) that combines spatial graph message passing with temporal processing using Leaky Integrate-and-Fire (LIF) spiking neurons. The temporal integration of membrane potentials is expected to help distinguish persistent patterns from isolated fluctuations, while the event-driven nature of spike-based computation can reduce unnecessary processing when activity is sparse.
OBJECTIVES OF THE PROJECT	• Design and implement a Temporal Spiking Graph Convolutional Network (TS-GCN) that combines spatial graph message passing using PyTorch Geometric with Leaky Integrate-and-Fire (LIF) spiking neuron dynamics for temporal processing. • Implement surrogate-gradient-based learning using the ArcTangent surrogate function • Develop a dynamic network transaction anomaly dataset pipeline that incorporates temporal graph changes and class-imbalance handling to represent realistic financial transaction and network anomaly scenarios. • Evaluate and compare TS-GCN with a conventional GCN baseline using classification accuracy, minority-class F1-score, ROC-AUC, spike sparsity, estimated neuromorphic energy consumption based on SOPs and FLOPs, and GPU training latency.
SUMMARY OF THE PROJECT	This project proposes a Temporal Spiking Graph Convolutional Network (TS-GCN), a bio-inspired deep learning framework for detecting anomalies in dynamic graphs, with a focus on financial transaction and network activity. The proposed system combines PyTorch Geometric for spatial graph message passing with snnTorch for temporal spiking neuron processing. The system is designed in four main stages: (1) generation of dynamic graph data representing changes in relationships and feature patterns associated with anomalous activity; (2) spatial graph message passing combined with Leaky Integrate-and-Fire (LIF) neuron dynamics to capture information across successive time steps; (3) surrogate-gradient-based training with membrane potential reset to enable learning in the spiking network; and (4) comparative evaluation of the proposed TS-GCN against a conventional GCN baseline. The proposed spiking approach is expected to produce sparse neural activity, allowing a significant portion of unnecessary computations to be avoided when graph activity is limited. The temporal integration

	provided by LIF neurons is also expected to improve the handling of short-term fluctuations in streaming data by retaining relevant information across time steps. Compared with a conventional GCN, the proposed model is therefore expected to show improved robustness to temporal noise, better detection of minority anomalous classes, lower computational activity, and improved energy efficiency, while maintaining competitive classification performance. The effectiveness of the proposed approach will be determined after implementation through comparative evaluation using classification accuracy, minority-class F1-score, ROC-AUC, spike sparsity, estimated energy consumption, and computational latency.
MODE OF CARRYING OUT THE PROJECT	• The project will be carried out within the Department of Information Science and Engineering, under the guidance of the project faculty. The development and evaluation will use open-source software frameworks and libraries required for graph learning, spiking neural networks, data processing, and model evaluation. • The proposed TS-GCN will be developed and evaluated through a GPU-accelerated software simulation environment. The implementation will use PyTorch-based tools along with PyTorch Geometric and snnTorch to simulate graph-based spiking neural network behaviour and study its computational characteristics. The project will focus on algorithmic and simulation-based evaluation rather than requiring direct access to dedicated neuromorphic hardware. • The development process will be carried out iteratively, covering dataset preparation, graph construction, model development, training, parameter tuning, and comparative evaluation. Regular reviews with the project guides will be conducted to assess progress and refine the implementation. • Git-based version control will be used to manage the source code and track development progress. The proposed TS-GCN will be evaluated against a conventional GCN baseline using a suitable train-test evaluation strategy and common performance measures such as classification accuracy, minority-class F1-score, ROC-AUC, spike sparsity, estimated energy consumption, and computational latency.
INTENDED BENEFICIARIES OF THE PROJECT	• **Financial Institutions and Payment Processors:** Organizations that monitor financial transaction networks, including card, banking, and digital payment transactions, could benefit from approaches that improve the detection of

	suspicious transaction patterns while reducing computational requirements for large-scale graph processing. • **Cybersecurity and Network Monitoring Teams:** Security teams monitoring enterprise, IoT, and other computer networks could use graph-based anomaly detection methods to identify unusual communication patterns and potentially detect network-related threats in continuously changing environments. • **Neuromorphic and Edge AI Researchers:** Researchers and developers working on energy-efficient AI can benefit from the study of combining graph neural networks with spiking neural network dynamics, particularly for applications where sparse and event-driven computation is desirable on neuromorphic or edge-computing platforms. • **Cloud and Data Center Operators:** Organizations processing large volumes of graph-structured data could potentially benefit from computationally efficient models that reduce unnecessary neural activity and make high-throughput graph processing more resource-efficient.
BASE PAPERS/ RELATED WORK	1. Eshraghian, J. K., Ward, M., Neftci, E. O., et al., "Training Spiking Neural Networks Using snnTorch," *Proceedings of the IEEE*, vol. 111, no. 9, pp. 1101–1116, 2023. Publisher: IEEE | Indexing: Web of Science 2. Neftci, E. O., Mostafa, H., and Indiveri, G., "Surrogate Gradient Learning Enabled Backpropagation for Spiking Neural Networks," *IEEE Signal Processing Magazine*, vol. 36, no. 6, pp. 51–63, 2019. Publisher: IEEE | Indexing: Web of Science 3. Tavanaei, A., Ghodrati, M., Kheradpisheh, S. R., Masquelier, T., and Maida, A., "Deep Learning in Spiking Neural Networks," *Neural Networks*, vol. 111, pp. 47–63, 2019. Publisher: Elsevier | Database: ScienceDirect 4. Kipf, T. N. and Welling, M., "Semi-Supervised Classification with Graph Convolutional Networks," *International Conference on Learning Representations (ICLR)*, 2017. Publisher: ICLR 5. Davies, M., Srinivasa, N., Lin, T. H., et al., "Loihi: A Neuromorphic Manycore Processor with On-Chip

	Learning," *IEEE Micro*, vol. 38, no. 1, pp. 82–99, 2018. Publisher: IEEE | Indexing: IEEE Xplore / ACM Digital Library
ABSTRACT	Real-time anomaly and fraud detection in dynamic networks requires models that can effectively capture changing relationships while keeping computational requirements manageable. Conventional Graph Neural Networks (GNNs) rely on continuous numerical operations for graph feature aggregation, which can become computationally expensive as the size and activity of the graph increase. They may also be sensitive to short-term fluctuations in streaming data when temporal information is not explicitly modelled. This project proposes TS-GCN (Temporal Spiking Graph Convolutional Network), a bio-inspired framework that combines spatial graph convolution using PyTorch Geometric with temporal processing through Leaky Integrate-and-Fire (LIF) spiking neuron dynamics. The proposed model will use surrogate-gradient-based learning and membrane potential reset mechanisms to enable effective training of the spiking network. The system will be developed to represent dynamic changes in transaction and network graphs and will be evaluated against a conventional non-spiking GCN baseline. The proposed approach is expected to exploit the sparse and event-driven nature of spiking computation, potentially reducing unnecessary neural activity and improving computational efficiency. The temporal integration provided by LIF neurons is also expected to help distinguish persistent patterns from isolated fluctuations in dynamic data, which may improve the detection of anomalous and fraudulent activity. The project will investigate these potential benefits through comparative evaluation of classification performance, minority-class detection, ROC-AUC, spike sparsity, estimated energy consumption, and computational latency.
KEYWORDS	Spiking Neural Networks; Graph Neural Networks; Temporal Graph Learning; TS-GCN; Neuromorphic Computing; Leaky Integrate-and-Fire; Surrogate Gradient Learning; Fraud Detection; Network Anomaly

	Detection; Spike Sparsity; PyTorch Geometric; snnTorch; Energy-Efficient AI.

Signature of the Guide	**Signature of the Project Coordinators**